

Descriptive Statistics Project

Name: Hanna [REDACTED]

Instructor: Samuel Chukwuemeka

Date: 11/18/2025

Objectives:

I shall do the following objectives:

- (1.) Dataset Description: Describe the dataset.
- (2.) Data Presentation: Present the dataset using a histogram.
- (3.) Measures of Center: Calculate and interpret the measures of center.
- (4.) Measures of Spread: Calculate and interpret the measures of spread.
- (5.) Measures of Position: Calculate and interpret the measures of position.

Description:

Descriptive Statistics Project: Monthly Total Precipitation for Richmond Area, VA from January, 2000 – December, 2000.

URL: <https://www.weather.gov/wrh/Climate?wfo=akq>

Navigation:

(1.) Location: Richmond Area, VA

(2.) Product: Monthly summarized data

(3.) Year range: 2000 - 2000;

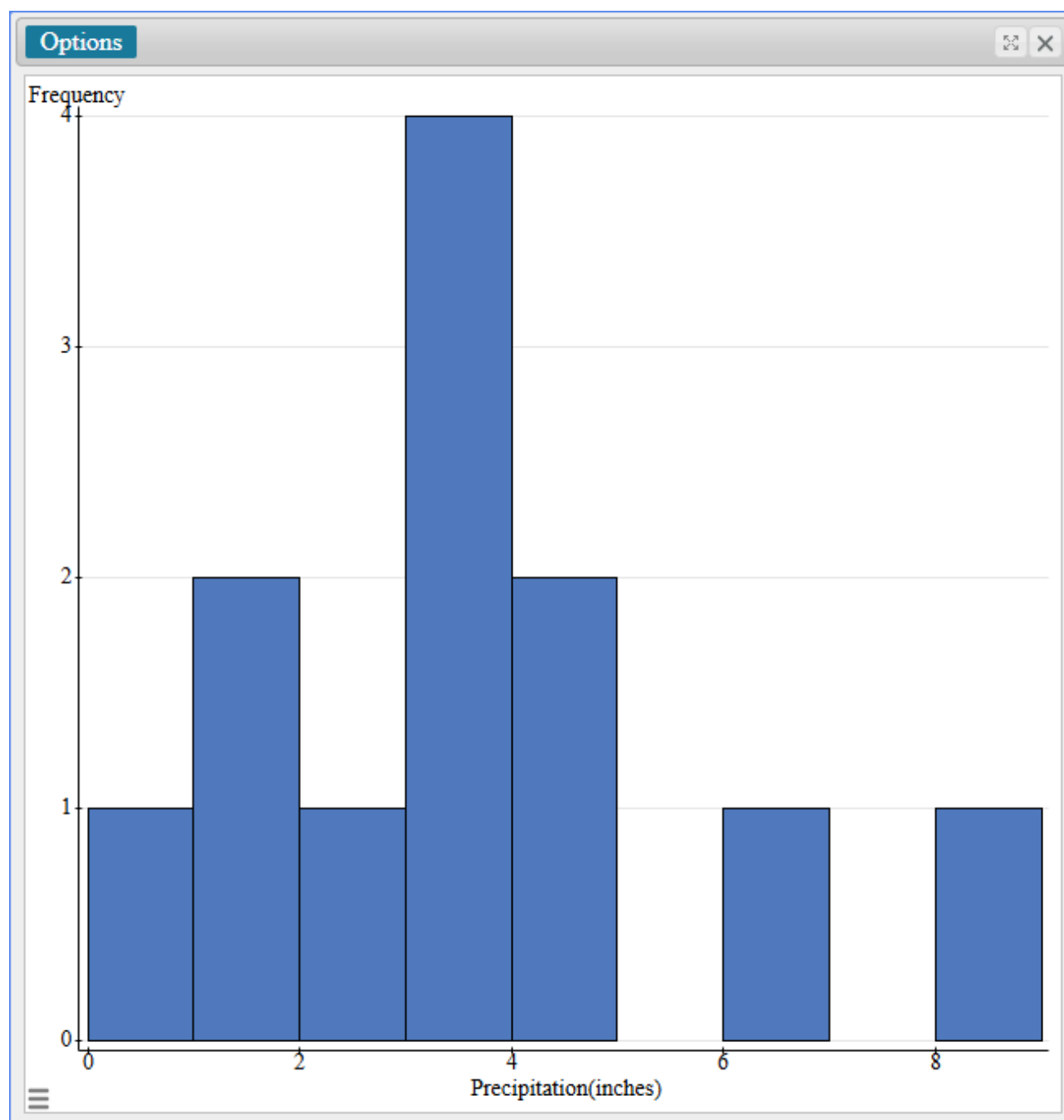
Variable: Precipitation;

Summary: Sum

(4.) View: Go

Should bring you to this data set:

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2000	3.96	1.60	3.67	4.78	3.03	6.07	4.05	8.28	3.60	0.01	1.72	2.38

Data Presentation:

Precipitation(in	Month
3.96	January
1.6	February
3.67	March
4.78	April
3.03	May
6.07	June
4.05	July
8.28	August
3.6	September
0.01	October
1.72	November
2.38	December

Notable Information:

[3, 4)

4 months have precipitation between 3 inches (included) and 4 inches (excluded).

Data Analysis:

Options

Summary statistics:

Column	Mode	n	Mean	Variance	Std. dev.	Median	Range	Min	Max	Q1	Q3	Sum	IQR
Precipitation	No mode	12	3.5958333	4.7243902	2.1735662	3.635	8.27	0.01	8.28	2.05	4.415	43.15	2.365

We shall use formulas, Pearson StatCrunch software, and Texas Instrument (TI-84 CE) calculator.

I shall round all final answers to **two decimal places if necessary**.

We shall begin with measures of center.

Measures of Center

3.96, 1.6, 3.67, 4.78, 3.03, 6.07, 4.05, 8.28, 3.6, 0.01, 1.72, 2.38

Sample size = 12

Mean:

$$\text{Mean} = \frac{\text{Sum of Data Values}}{\text{Sample Size}} = \frac{43.15}{12} = 3.5958333 \approx 3.6$$

The average monthly total precipitation for Richmond Area, VA from January, 2000 – December, 2000 is approximately 3.6 inches.

Median:

I am going to sort the data in ascending order:

0.01, 1.6, 1.72, 2.38, 3.03, 3.6, 3.67, 3.96, 4.05, 4.78, 6.07, 8.28

$$\text{Median} = Q_2 = \frac{3.6 + 3.67}{2} = \frac{7.27}{2} = 3.635 \approx 3.64$$

When the monthly total precipitation for Richmond Area, VA from January, 2000 to December, 2000 is in ascending order, the median value is approximately 3.64 inches.

Mode:

In the dataset, there is no precipitation value that occurred more frequently than any other value.

Midrange:

$$\text{Minimum} = 0.01$$

$$\text{Maximum} = 8.28$$

$$\text{Midrange} = \frac{\text{minimum} + \text{maximum}}{2} = \frac{0.01 + 8.28}{2} = \frac{8.29}{2} = 4.145 \approx 4.15$$

During January, 2000 – December, 2000 for Richmond, VA the average precipitation of the minimum and maximum precipitation values is approximately 4.15 inches.

Measures of Spread

Range:

$$\text{Minimum} = 0.01$$

$$\text{Maximum} = 8.28$$

$$\text{Range} = \text{maximum} - \text{minimum} = 8.28 - 0.01 = 8.27$$

The difference between the maximum and the minimum values for monthly total precipitation for Richmond, VA from January, 2000 to December, 2000 is 8.27 inches.

Variance and Standard Deviation:

I shall round all intermediate values to 5 decimal places as needed.

<i>Precipitation, x</i>	<i>$x - \text{Mean}$</i>	<i>$(x - \text{Mean})^2$</i>
3.96	0.36417	0.13262
1.6	−1.99583	3.98335
3.67	0.07417	0.00550
4.78	1.18417	1.40225
3.03	−0.56583	0.32017

6.07	2.47417	6.12150
4.05	0.45417	0.20627
8.28	4.68417	21.94142
3.6	0.00417	1.73611
0.01	−3.58583	12.85820
1.72	−1.87583	3.51875
2.38	−1.21583	1.47825
$\Sigma(x - \text{Mean})^2 = 51.96829167$		

$$\text{Variance} = \frac{\Sigma(x - \text{Mean})^2}{n - 1} = \frac{51.96829167}{12 - 1}$$

$$= 4.724390152 \approx 4.72 \text{ square inches}$$

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

$$= \sqrt{4.724390152} = 2.173566229 \approx 2.17 \text{ inches}$$

The value of the standard deviation shows there is considerable spread among the precipitation values.

Interquartile Range (IQR):

The dataset arranged in ascending order:

0.01, 1.6, 1.72, 2.38, 3.03, 3.6, 3.67, 3.96, 4.05, 4.78, 6.07, 8.28

$$n = 12$$

$$xth\ position = \frac{xth\ position + (x + 1)th\ position}{2} \text{ (for integer)}$$

$$\frac{1}{4} * 12 = 3$$

Since 3 is an integer, Q_1 is the average of the 3rd and 4th positions.

$$3rd\ position = 1.72$$

$$4th\ position = 2.38$$

$$Q_1 = \frac{1.72 + 2.38}{2} = \frac{4.1}{2} = 2.05$$

$$\frac{3}{4} * 12 = 9$$

Since 9 is an integer, Q_3 is the average of the 9th and 10th positions.

$$9th\ position = 4.05$$

$$10th\ position = 4.78$$

$$Q_3 = \frac{4.05 + 4.78}{2} = \frac{8.83}{2} = 4.415$$

$$IQR = Q_3 - Q_1 = 4.415 - 2.05 = 2.365$$

The variation in 50% of the monthly total precipitation for Richmond Area, VA from January, 2000 – December, 2000 is approximately 2.365 inches.

Measures of Position

I shall use the five-number summary of data.

$$\textit{Minimum} = 0.01$$

The minimum value of the monthly total precipitation for Richmond Area, VA from January, 2000 – December, 2000 is 0.01 inches.

$$\textit{First Quartile}, Q_1 = 2.05$$

25% of the monthly total precipitation for Richmond Area, VA from January, 2000 – December, 2000 is below 2.05 inches.

Middle Quartile, Q_2 (Median) = 3.635

3.635 inches is in the 50th percentile of monthly total precipitation for Richmond Area, VA from January, 2000 – December, 2000.

Upper Quartile, Q_3 = 4.415

25% of the monthly total precipitation for Richmond Area, VA from January, 2000 – December, 2000 is above 4.415 inches.

Maximum = 8.28

The maximum value of the monthly total precipitation for Richmond Area, VA from January, 2000 – December, 2000 is 8.28 inches.

Reference Page (MLA 9 Style)

US Department of Commerce, NOAA. "Climate." *Www.weather.gov*, 2000,

www.weather.gov/wrh/Climate?wfo=akq.

Chukwuemeka, Samuel. "Descriptive Statistics." *Descriptive Statistics Projects*,
conferencepresentations.appspot.com/Projects/Statistics/DescriptiveStatistic
s.html. Accessed 18 Nov. 2025.